

Tables



1 $\frac{5}{36}$

2 a Sample Space = {0, 1, 2, 3, 4, 5}

b $\frac{1}{6}$

3 a

Spinner	2	3	4
7	9	10	11
9	11	12	13
12	14	15	16

b i $\frac{2}{9}$ ii $\frac{2}{9}$ iii $\frac{3}{9}$

4 a

	1	2	3	4	5	6
1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

b i $\frac{1}{18}$ ii $\frac{1}{36}$ iii $\frac{1}{6}$ iv $\frac{1}{2}$

5 a

	W	X	Y	Z
1	1, W	1, X	1, Y	1, Z
2	2, W	2, X	2, Y	2, Z
3	3, W	3, X	3, Y	3, Z
4	4, W	4, X	4, Y	4, Z
5	5, W	5, X	5, Y	5, Z
6	6, W	6, X	6, Y	6, Z

b 24
c i $\frac{1}{8}$ ii $\frac{1}{6}$ iii $\frac{11}{12}$



6

a

	8	2	3	5
1	8	2	3	5
2	16	4	6	10
3	24	6	9	15
4	32	8	12	20
5	40	10	15	25
6	48	12	18	30

b

$$\frac{1}{4}$$

c

$$\frac{1}{12}$$

d

$$\frac{2}{3}$$

7

a

	Cyan (<i>C</i>)	Pink (<i>P</i>)	White (<i>W</i>)
Black (<i>B</i>)	<i>C, B</i>	<i>P, B</i>	<i>W, B</i>
Grey (<i>G</i>)	<i>C, G</i>	<i>P, G</i>	<i>W, G</i>
Red (<i>R</i>)	<i>C, R</i>	<i>P, R</i>	<i>W, R</i>
Yellow (<i>Y</i>)	<i>C, Y</i>	<i>P, Y</i>	<i>W, Y</i>

b

12

c

$$\frac{1}{4}$$

d

$$\frac{1}{3}$$

8

a

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	25	41	4	70
Liberal (last election)	16	73	1	90
Greens (last election)	9	26	5	40
Total	50	140	10	200

b

$$\frac{8}{45}$$

c

$$\frac{1}{2}$$

d $\frac{57}{80}$



9

a

Spinner	2	4	5
3	5	7	8
7	9	11	12

b

- i 1 ii $\frac{5}{6}$ iii $\frac{2}{5}$ iv 1

Tree diagrams

10 $\frac{1}{9}$

11

a SR, SL, SC, GV, GM, GC

b $\frac{1}{6}$

c $\frac{1}{3}$

12

a {2, 3, 5, 7, 11, 13, 17, 19}

- b i $\frac{1}{8}$ ii $\frac{1}{4}$

13

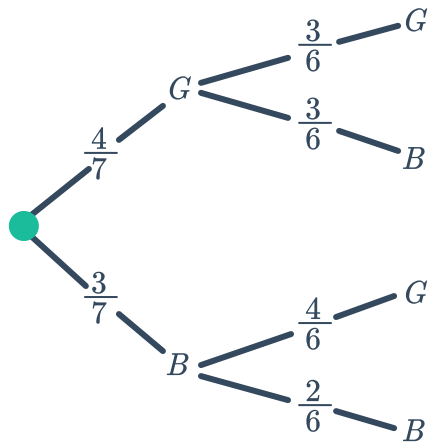
a 23, 24, 32, 34, 42, 43

- b i $\frac{2}{3}$ ii $\frac{1}{3}$ iii $\frac{1}{3}$

14

a 0.42 b 0.88

15 a

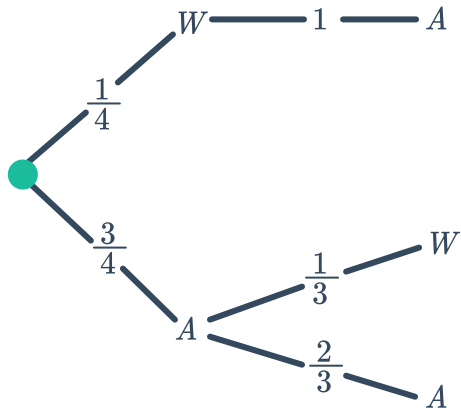


b

i $\frac{3}{7}$

ii $\frac{2}{7}$

16 a

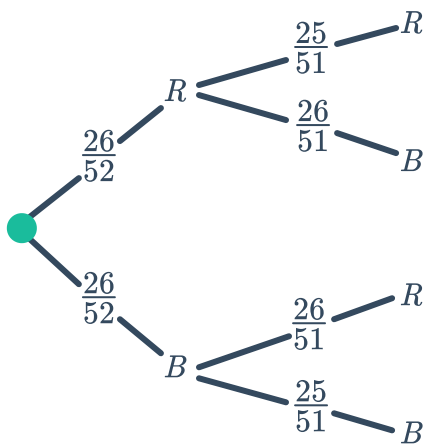


b

i $\frac{3}{4}$

ii 1

17 a



b $\frac{13}{51}$

c

i Yes

ii No

iii No

iv No

v No

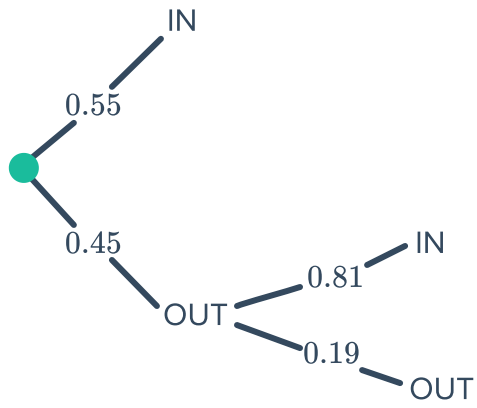
18 a



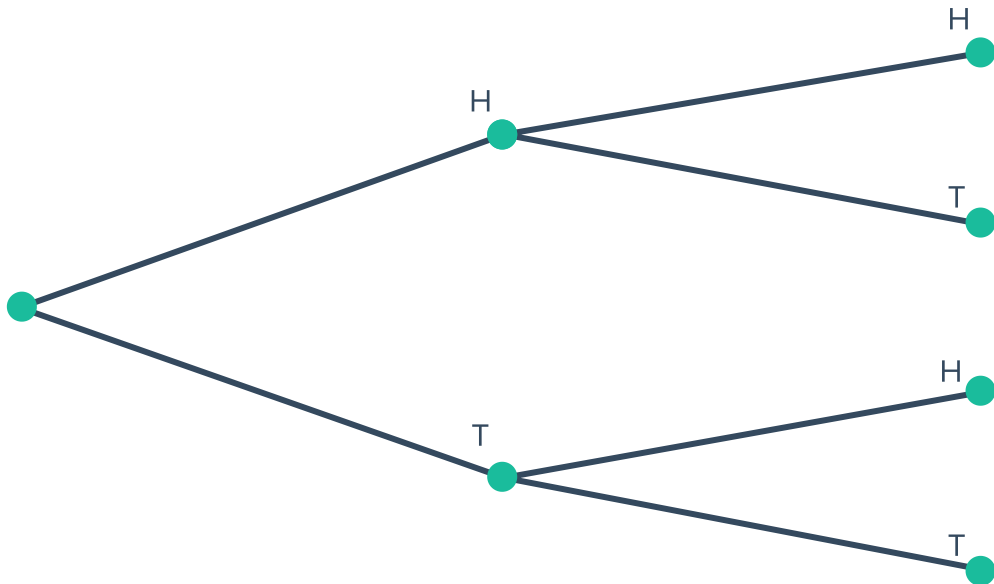
b

i 0.45

ii 0.0855



19 a



b

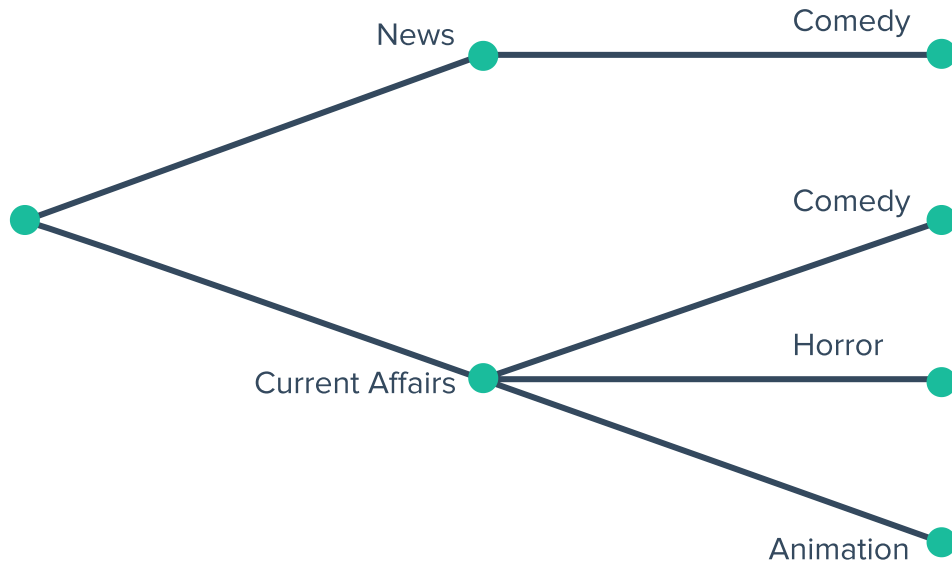
i $\frac{2}{4}$

ii $\frac{1}{4}$

iii $\frac{1}{4}$

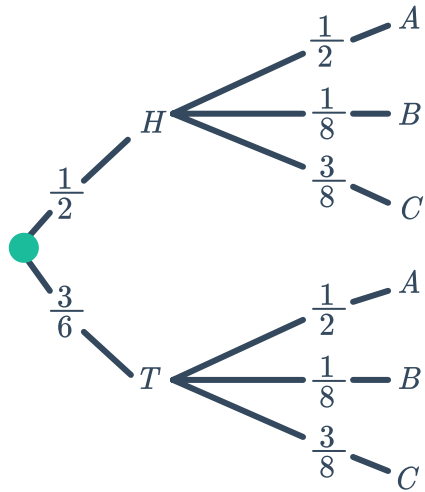
iv $\frac{2}{4}$

20
a



b $\frac{2}{3}$

21
a



b i $\frac{3}{16}$

ii $\frac{11}{16}$

22 $\frac{1}{7776}$

23



a

i $\frac{1}{9}$

ii $\frac{1}{27}$

iii $\frac{19}{27}$

b

$P(\text{Winning at least one of two games}) = \frac{5}{9}$, $P(\text{Winning at least one of three games}) = \frac{19}{27}$
therefore Neville has a better chance at winning at least one of three games.

24

a

i $\frac{1}{2}$

ii $\frac{1}{4}$

iii $\frac{1}{8}$

b $\frac{1}{2}$

c

$\frac{1}{16}$

25

a

$\frac{1}{2}$

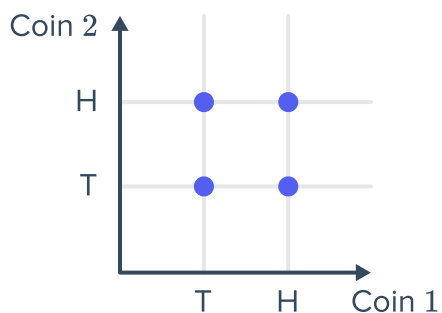
b

$\frac{1}{32}$

2D Grids

26

a



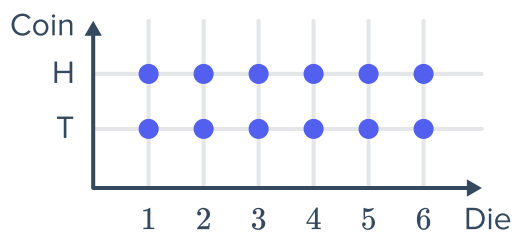
b 4

c

$\frac{1}{2}$

27

a



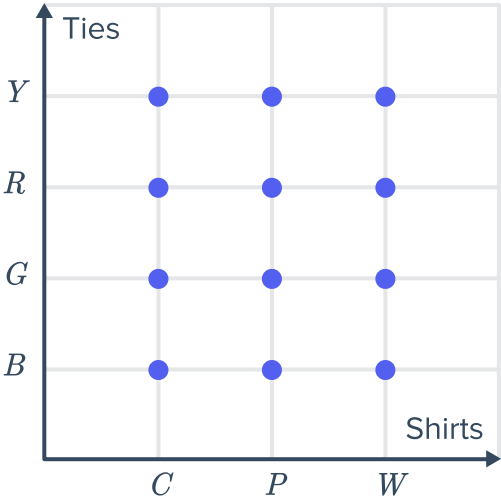
b 12

28 a

	<i>C</i>	<i>P</i>	<i>W</i>
<i>B</i>	<i>C, B</i>	<i>P, B</i>	<i>W, B</i>
<i>G</i>	<i>C, G</i>	<i>P, G</i>	<i>G, W</i>
<i>R</i>	<i>C, R</i>	<i>P, R</i>	<i>W, R</i>
<i>Y</i>	<i>C, Y</i>	<i>P, Y</i>	<i>W, Y</i>



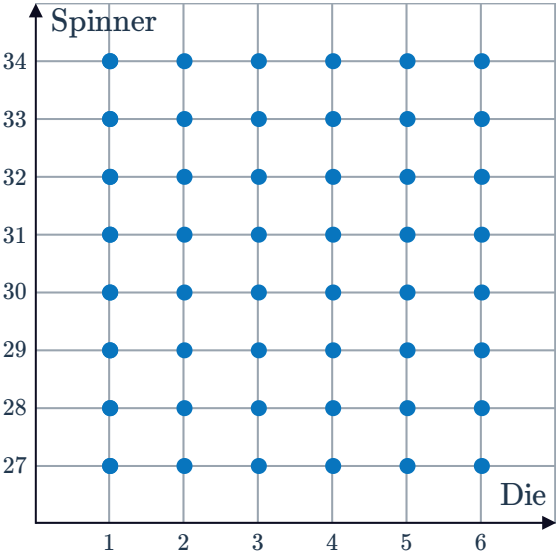
b



c 12

29

a



b 48

c $\frac{1}{4}$